

SMART WARI AI

An AI Coordination Layer for Pilgrimage Safety

“We don't ask lakhs of pilgrims to change their behavior — we make the existing cameras and phone lines smart.”

Track	Open Innovation Track (see note on track alignment, p. 2)
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1. Problem Statement

Every year the Pandharpur Wari brings together 10–20 lakh pilgrims walking hundreds of kilometres across Maharashtra. Despite extensive planning by police and local authorities, three operational challenges recur every season:

- **Crowd congestion:** Rapid, unpredictable buildup at bottlenecks creates stampede risk.
- **Missing persons:** Elderly pilgrims and children frequently get separated from their groups and are difficult to locate.
- **Delayed medical response:** Heatstroke, dehydration, and crowd-related injuries are common, and response times are slow.

Existing technology solutions (apps, QR bands, wearables, kiosks) largely fail here because most pilgrims do not own smartphones, are not digitally literate, cannot be expected to install an app mid-journey, and are constantly on the move. Meanwhile, thousands of CCTV cameras, police control rooms, a volunteer network, and helpline numbers already exist — but operate in silos with no shared intelligence.

Core insight: the gap is not infrastructure — it is the absence of an intelligence layer connecting the infrastructure that already exists.

2. Project Idea

Smart Wari AI is a software-only AI coordination layer that sits above existing CCTV cameras, police control rooms, toll-free phone lines, and the volunteer network — requiring no new hardware, apps, or wearables for pilgrims. It continuously analyses live CCTV streams with computer vision while simultaneously processing incoming pilgrim voice calls, turning today's disconnected surveillance feeds into a real-time operational intelligence platform for Wari organisers.

- **Zero-adoption for pilgrims:** no app, no QR band, no wearable, no registration — pilgrims behave exactly as they do today.
- **Built on what already exists:** CCTV network, toll-free IVR line, police control room, on-ground volunteers.
- **Software-only, cloud-deployable:** low cost, fast to pilot, and scalable to any large public gathering.

3. Proposed Solution

The platform consists of three integrated AI engines feeding a single Unified Coordination Engine and Command Center Dashboard.

3.1 Crowd Intelligence Engine

Purpose: predict dangerous congestion before it happens. CCTV feeds are continuously analysed for crowd density, walking speed, direction of movement, and flow bottlenecks. Results are rendered as a live colour-coded heatmap:

Green – Normal | Yellow – Moderate crowd | Orange – High density | Red – Critical congestion

Based on the heatmap, the system suggests concrete actions to organisers — deploy a water tanker, relocate food distribution, open an alternate route, or increase police deployment — turning passive recording cameras into a predictive crowd-management tool.

3.2 AI Voice-based Lost & Found

Instead of installing an app, a pilgrim (or a bystander) simply calls a toll-free number and describes the missing person in their own language, e.g. “माझे बाबा हरवले आहेत. नाव विठ्ठल पाटील. त्यांनी पांढरा कुर्ता घातला आहे.” The system runs this call through a fully automated pipeline:

Speech → Speech-to-Text → Language Understanding → Extract name / age / gender / clothing → Generate a search profile → Search the CCTV network → Identify the last-seen camera → Push the location to the Police Control Room → Nearest volunteer receives navigation instructions.

No typing, no smartphone, no QR code, and no prior registration are required — only a phone call.

3.3 Medical Risk Detection

The same CCTV pipeline is reused to flag emergencies in real time: a person collapsing, a crowd suddenly clustering around someone, abnormal inactivity, a sudden fall, or slow movement suggesting exhaustion. This is fused with live weather data (temperature, humidity) and crowd density to predict Heat Risk Zones. Medical volunteers then receive targeted alerts such as “Possible medical emergency near Camera 42,” cutting response time significantly.

3.4 Unified Command Center

All three engines converge into one dashboard for the police control room: live camera grid, crowd heatmap, lost-person desk, medical alerts, resource status (water tankers, food vans, volunteers), and route/diversion status — everything visible in one place, enabling data-driven, predictive rather than reactive decision-making.

4. System Architecture

The architecture layers cleanly from existing infrastructure at the bottom to decision-makers at the top: Existing CCTV + Toll-Free IVR → Stream/Speech Processing → AI Processing Layer (Crowd Intelligence, Lost & Found, Medical Risk) → Smart Decision / Unified Coordination Engine → Command Center Dashboard → Police, Field Volunteers, and Medical Teams.

SMART WARI AI

AI Coordination Layer over Existing Infrastructure

"We don't ask lakhs of pilgrims to change behavior -- we make the existing cameras and phone lines smart."

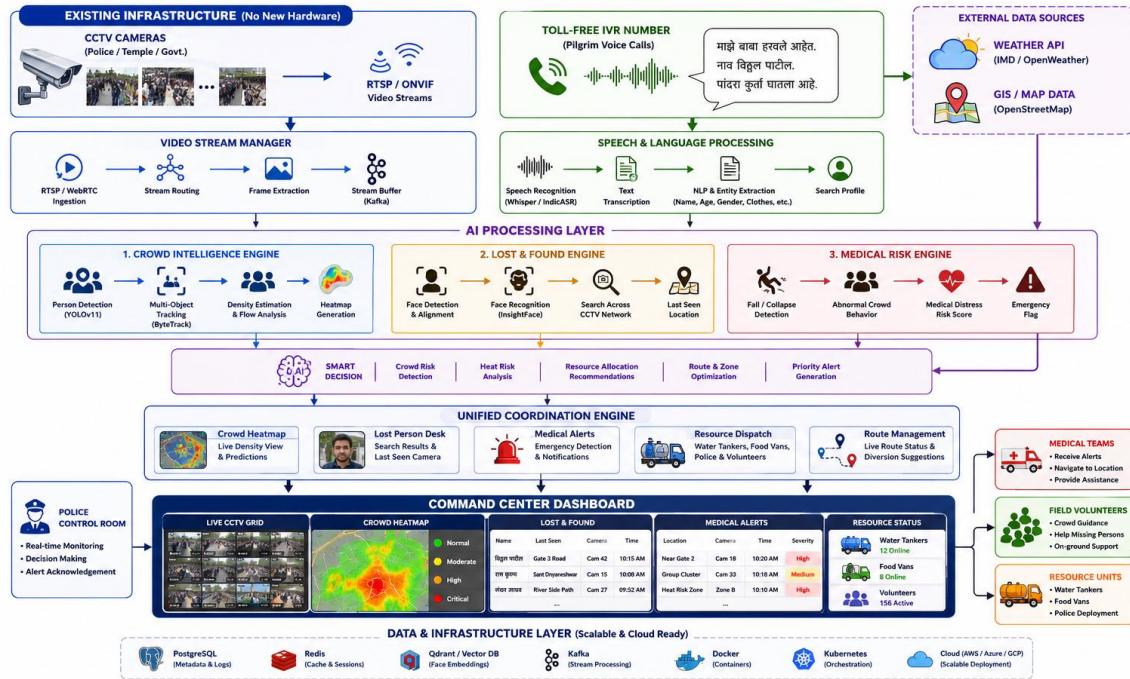


Figure 1: Smart Wari AI – end-to-end system architecture

5. Complete Tech Stack

Layer	Technologies
Frontend	React.js, Next.js, Tailwind CSS, MapLibre GL, Chart.js
Backend	FastAPI, Python, Node.js (notification services)
Computer Vision	YOLOv11, ByteTrack / DeepSORT, OpenCV, TensorFlow / PyTorch
Face Recognition	InsightFace, FaceNet, ArcFace
Speech AI	OpenAI Whisper, IndicWhisper, Marathi ASR
Language Processing	Indic NLP, Llama 3 / Gemma (optional)
Forecasting	LSTM, Prophet, XGBoost
Databases	PostgreSQL, Redis, Qdrant (vector DB)
Streaming	RTSP, WebRTC, Kafka, Redis Streams
GIS	OpenStreetMap, MapLibre
External APIs	IMD Weather API, OpenWeather API, SMS Gateway, IVR (Exotel / Twilio)
Deployment	Docker, Kubernetes, NGINX, AWS / Azure / GCP

6. Implementation Roadmap

A six-phase roadmap covers development through a live, deployed pilot — sized to fit incremental delivery across Stage 1 report, Stage 2 demo video, and the Stage 3 on-site hackathon build.

Phase	Focus	Key Activities	Output
Phase 1	Infrastructure Integration	Connect RTSP CCTV (or simulated) feeds, build streaming pipeline, store camera metadata, build base dashboard shell.	Live CCTV monitoring view
Phase 2	Crowd Intelligence	YOLO person detection → tracking → density estimation → heatmap → prediction model → resource recommendation.	Live crowd dashboard
Phase 3	Voice Lost & Found	Call ingestion → speech recognition → Marathi NLP → extract details → face search → last-seen location → notify police.	Complete lost-person workflow
Phase 4	Medical Emergency Detection	Detect fall / collapse / crowd clustering → fuse weather risk → compute risk score → notify volunteers.	Medical alert system
Phase 5	Unified Command Center	Integrate all engines into one dashboard: cameras, heatmap, lost persons, medical alerts, volunteers, resources, routes.	Single command center
Phase 6	Deployment & Pilot	Containerise services, deploy frontend + backend to cloud, wire a real IVR number, connect simulated CCTV feed, load-test, share live URL.	Publicly accessible live demo

7. Deployment Strategy (Live, Working Demo)

Since the hackathon cannot connect to real police CCTV feeds, the deployed demo uses the identical pipeline against a substitute video source so every component runs live rather than as a mock-up:

- **Simulated CCTV input:** 2–3 pre-recorded / looped crowd videos (public dataset or self-shot footage) served over RTSP using an open-source media server (e.g. MediaMTX) to stand in for real camera feeds — the AI pipeline treats it exactly like a live stream.
- **Frontend:** React/Next.js dashboard deployed on Vercel, connected to the backend via secure WebSocket/REST.
- **Backend & AI services:** FastAPI services containerised with Docker and deployed on a cloud VM or container platform (Railway / Render / AWS ECS) — one container for detection & tracking, one for the speech pipeline, one for the API/orchestration layer.
- **Voice line:** A real toll-free/virtual number via Exotel or Twilio IVR, wired to the speech-to-text + NLP pipeline, so judges can literally call in a test “missing person” report.
- **Data layer:** Managed PostgreSQL + Redis (e.g. Supabase/Neon + Upstash) for metadata, sessions, and the vector index (Qdrant Cloud free tier) for face-embedding search.
- **Domain & access:** a single public URL for the Command Center Dashboard, shared with judges for live, hands-on evaluation rather than a recorded walkthrough.

This lets the team demonstrate an actually-running command center — live heatmap updating on the simulated feed, a real inbound call resolving to a “last seen” camera, and a medical alert firing — within the constraints of a 24-hour on-site hackathon.

8. Feasibility

The project is feasible primarily because it adds no new physical dependency — no manufacturing, sensor installation, or hardware logistics are required:

- CCTV infrastructure already exists across police, temple trust, and municipal networks.
- RTSP/ONVIF streaming is supported by most existing surveillance systems.
- Marathi and Indic speech-recognition models (Whisper, IndicWhisper) are available and improving rapidly.
- Open-source computer-vision models (YOLOv11, ByteTrack) achieve strong real-time accuracy.
- Face-recognition libraries (InsightFace, ArcFace) are mature and production-tested.
- Weather APIs (IMD, OpenWeather) are freely available.
- IVR providers (Exotel, Twilio) connect phone calls without dedicated telecom hardware.
- The entire solution is software-based and cloud-deployable end to end.

Viability Snapshot

Factor	Assessment
Deployment Cost	Low — no new hardware, cloud-metered compute only
Maintenance	Software updates only
Hardware Requirement	None — reuses existing CCTV and telecom lines
Pilot Deployment	Possible on a small subset of cameras before full-scale rollout

9. Target Customers & Business Model

Target Customers

- Maharashtra Police
- District Administration
- Disaster Management Authorities
- Temple Trusts
- Municipal Corporations
- Large event organisers

Business Model

- Software-as-a-Service (SaaS) licensed to government bodies.
- Annual maintenance and support contracts.
- Per-control-room subscription for dashboard access.

10. Impact

Social Impact

- Faster missing-person recovery

- Reduced stampede risk
- Improved emergency response time
- Better crowd flow management
- Fully inclusive for digitally illiterate or non-smartphone pilgrims

Economic Impact

- No new hardware purchases — uses existing camera and telecom investments efficiently.
- Lowers ongoing operational cost of manual crowd monitoring.

Administrative Impact

- Single command center for all decision-making.
- Data-driven, predictive rather than reactive management.

11. Scalability

The same software stack — only the CCTV feed source changes — can be redeployed for other large gatherings across India, including: Kumbh Mela, Ganesh Visarjan, Jagannath Rath Yatra, Sabarimala Pilgrimage, Mahakaleshwar festivals, political rallies, cricket stadiums, music festivals, disaster relief camps, election crowd management, and railway/metro/airport terminals.

12. Future Scope

- Digital twin simulation of Wari routes for pre-event planning.
- Multilingual voice support: Hindi, Kannada, Telugu, English.
- AI-optimised volunteer assignment and dispatch.
- Predictive crowd simulation several hours ahead of buildup.
- Anonymous, privacy-preserving crowd analytics for policy planning.
- Integration with emergency ambulance routing.
- Optional drone video integration where available.
- AI-generated daily operational reports for authorities.
- Federated learning for privacy-preserving model improvement across deployments.

13. Research & Reference Resources

Area	Tools / References
Computer Vision	YOLOv11, ByteTrack, DeepSORT, OpenCV, InsightFace, ArcFace
Speech	Whisper, IndicWhisper, Vosk
NLP	Indic NLP Library, spaCy, Llama 3 / Gemma
Streaming	RTSP, WebRTC, Kafka, Redis Streams
GIS	OpenStreetMap, MapLibre

Area	Tools / References
Databases	PostgreSQL, Redis, Qdrant
Cloud	Docker, Kubernetes, AWS / Azure / GCP

14. Alignment with Varithon 2026

Varithon 2026 (E&TC Department, MMCOE Pune) is built around solving real structural and logistical challenges of the Pandharpur Wari — crowd management, healthcare accessibility, and emergency deployment are named explicitly in the problem brief. Smart Wari AI addresses all three directly, and does so by building on infrastructure that authorities already operate, matching the event's call for “practical, high-throughput, scalable solutions.”

Note on track selection: the source material labels this an Open Innovation entry, but its core is crowd prediction (Track 2: Crowd, Mobility & Resource Management) and emergency/medical response (Track 1: Health, Safety & Emergency Response). Confirm which single track to submit under — see the questions at the end of this document.

15. Hackathon Pitch Summary

Smart Wari AI is not another surveillance system — it is an intelligence layer over infrastructure that already exists. Instead of asking millions of pilgrims to download apps, wear devices, or learn new technology, it empowers organisers with AI-driven coordination using the cameras and phone lines they already have. This zero-adoption, software-only approach makes the solution affordable, rapidly deployable, and scalable from the Wari to every large public gathering in India.